Family-macro Pearson r

Family-macro Kendall τb

Family-macro log10MAE

Family-macro RSE

TIGER-pair
P + G
P + S
S + G
P only
G only
S only
GAT
GraphSAGE
GCN
+ Energy
+ Coordinates
+ Energy + coordinates
+ Learned embedding
Zero node features
Sequence-chain edges
Cross-attention: P query
Cross-attention: G query
Feature concatenation
Self-attention: G/S/P
Self-attention: S/G/P
Concatenated pair head
Single-peptide regression
No training gate
Similarity ≥ 0.0
Similarity ≥ 0.3
Similarity ≥ 0.5
Similarity ≥ 0.9
Signed quota 1000
Signed quota 2000
Signed quota 4000
Signed quota 8000
Signed quota 10000
Signed quota 20000
Unsigned quota 1000
Unsigned quota 2000
Unsigned quota 4000
Unsigned quota 8000
Unsigned quota 10000
Negative-pair strategy: top-k
Strict within-fold validation
P00: neither stage

0.0 0.2 0.4 0.6

0.0 0.2 0.4

0.40 0.45 0.50

0.8 1.0

Mean ± sample SD across seeds 120–129. Dashed line: TIGER-pair (unsigned quota 20,000).